Table 1: All locally recorded JetNet-30 campaign results. Lower is better for FPND, FPD, W1M, mean W1P, and MMD; higher is better for coverage. Dashes mean no completed measurement. † marks historical, pre-canonical-FPND values and must not be compared to the literature. ‡ marks the GQT full-sample FPD, which is quarantined (raw value 646,050) pending a tail diagnostic.

Run	Design	Steps	Params.	FPND	FPD	W1M	W1P	Cov.	MMD	Physicality
A	Euclidean; no refs	200K	442K	—	—	0.2499	—	—	—	4.49% invalid; 2.29% neg.-E; 25.5% spacelike
B	RFM; no refs	200K	442K	91.7956†	0.1031	0.0257	0.0119	0.436	0.0301	0 invalid
C	Euclidean; plain refs	200K	451K	2.166×10^{27}†	762.9267	0.1057	0.0501	0.334	0.0451	0.006% invalid; 0.298% neg.-E; 33.9% spacelike
D	RFM; plain refs	200K	451K	16.3088†	221.1453	0.0089	0.0127	0.426	0.0301	0.004% invalid
E	RFM; scalar init + ICP; no ref readout	200K	442K	28.4260†	0.0525	0.0253	0.0165	0.389	0.0321	0 invalid
F	RFM; scalar init + ICP; plain refs	200K	451K	23.0223†	0.0253	0.0231	0.0152	0.405	0.0294	0 invalid
G-200K	Raw refs; evolving geometry	200K	617K	0.3935	1.593×10^{-4}	0.0019	0.0014	0.563	0.0265	0 invalid
H-200K	Normalized tangent refs; evolving geometry	200K	617K	0.3745	1.469×10^{-4}	0.0018	0.0013	0.565	0.0268	0 invalid
I-200K	Raw refs; fixed physical geometry	200K	562K	0.4264	1.685×10^{-4}	0.0020	0.0014	0.545	0.0279	0 invalid
J-200K	Normalized tangent refs; fixed physical geometry	200K	562K	0.4563	1.747×10^{-4}	0.0022	0.0015	0.539	0.0282	0 invalid
H-BlockCond-200K	H + condition/time in every block	200K	783K	0.3868	1.524×10^{-4}	0.0022	0.0013	0.544	0.0274	0 invalid
H-500K	H recipe extended	500K	617K	0.3678	1.397×10^{-4}	0.0020	0.0013	0.551	0.0284	0 invalid
G-500K	G recipe extended	500K	617K	0.3754	1.389×10^{-4}	0.0019	0.0014	0.555	0.0268	0 invalid
G-331K	H recipe; gluon specialist	331K	617K	0.3931	1.508×10^{-4}	0.0021	0.0014	0.536	0.0276	0 invalid
T-331K	H recipe; top specialist	331K	617K	—	0.4290	0.0058	0.0037	0.574	0.0566	0 invalid
GQT-994K (g/q/t)	H recipe; joint g/q/t	994K	617K	g=0.1311 q=0.2616 t=3.1530	—‡	0.0011	9.188×10^{-4}	0.562	0.0361	0.004% invalid

Table 2: Gluon-30 comparison. Values for MPGAN and PC-JeDi are quoted from their published gluon-30 tables; FPD was not reported there. All JetFUEL rows use the corrected canonical FPND input.

Method	Protocol / variant	FPND-g	FPD	W1M	Mean W1P	Cov.	MMD	Physicality
JetNet MC floor	not a trained generator; truth reference	0.0100	—	4.400×10^{-4}	4.100×10^{-4}	0.550	0.0367	—
MPGAN	2000 epochs; batch 256; jet attributes from test set	0.1200	—	7.000×10^{-4}	9.000×10^{-4}	0.560	0.0370	—
PC-JeDi DDIM	3000 epochs maximum; batch 256; jet attributes from test set	0.1200	—	9.000×10^{-4}	6.600×10^{-4}	0.540	0.0364	—
PC-JeDi EM	3000 epochs maximum; batch 256; jet attributes from test set	0.1000	—	5.700×10^{-4}	5.800×10^{-4}	0.540	0.0362	—
EPiC-FM*	10000 epochs; batch 1024; not gluon-30	—	—	—	—	—	—	—
EPiC-JeDi*	10000 epochs; batch 1024; not gluon-30	—	—	—	—	—	—	—
JetFUEL G-200K	Raw refs; evolving geometry	0.3935	1.593×10^{-4}	0.0019	0.0014	0.563	0.0265	0 invalid
JetFUEL H-200K	Normalized tangent refs; evolving geometry	0.3745	1.469×10^{-4}	0.0018	0.0013	0.565	0.0268	0 invalid
JetFUEL I-200K	Raw refs; fixed physical geometry	0.4264	1.685×10^{-4}	0.0020	0.0014	0.545	0.0279	0 invalid
JetFUEL J-200K	Normalized tangent refs; fixed physical geometry	0.4563	1.747×10^{-4}	0.0022	0.0015	0.539	0.0282	0 invalid
JetFUEL H-BlockCond-200K	H + condition/time in every block	0.3868	1.524×10^{-4}	0.0022	0.0013	0.544	0.0274	0 invalid
JetFUEL H-500K	H recipe extended	0.3678	1.397×10^{-4}	0.0020	0.0013	0.551	0.0284	0 invalid
JetFUEL G-500K	G recipe extended	0.3754	1.389×10^{-4}	0.0019	0.0014	0.555	0.0268	0 invalid
JetFUEL G-331K	H recipe; gluon specialist	0.3931	1.508×10^{-4}	0.0021	0.0014	0.536	0.0276	0 invalid
JetFUEL GQT-994K[g]‡	joint g/q/t; g subset FPND	0.1311	—	—	—	—	—	0.004% invalid

* EPiC-FM and EPiC-JeDi report top-30/top-150 jets, not gluon-30; their cells are intentionally blank and they are not a numerical comparison.
‡ GQT-994K is joint g/q/t training. Its FPND-g is class-filtered and comparable; its other saved metrics are joint-sample aggregates, so they are intentionally blank here. Its raw full-sample FPD is quarantined.
PC-JeDi and MPGAN condition on jet attributes taken from the test set, while JetFUEL samples those attributes with Stage 1; this is an important remaining protocol difference.

Table 3: Class-resolved JetNet-30 comparison. PC-JeDi reports gluon and top but not light-quark jets; MPGAN reports all three. FPD is excluded because the cited papers do not report it.

Class	Method	Protocol / variant	FPND	W1M	Mean W1P	Cov.	MMD	Physicality
g	JetNet MC floor	not a trained generator; truth reference	0.0100	4.400×10^{-4}	4.100×10^{-4}	0.550	0.0367	—
g	MPGAN	2000 epochs; batch 256; jet attributes from test set	0.1200	7.000×10^{-4}	9.000×10^{-4}	0.560	0.0370	—
g	PC-JeDi DDIM	3000 epochs maximum; batch 256; jet attributes from test set	0.1200	9.000×10^{-4}	6.600×10^{-4}	0.540	0.0364	—
g	PC-JeDi EM	3000 epochs maximum; batch 256; jet attributes from test set	0.1000	5.700×10^{-4}	5.800×10^{-4}	0.540	0.0362	—
q	MPGAN	2000 epochs; batch 256; jet attributes from test set	0.3500	6.000×10^{-4}	0.0049	0.500	0.0260	—
t	JetNet MC floor	not a trained generator; truth reference	0.0100	3.200×10^{-4}	4.000×10^{-4}	0.590	0.0713	—
t	MPGAN	2000 epochs; batch 256; jet attributes from test set	0.3600	6.400×10^{-4}	0.0022	0.580	0.0711	—
t	PC-JeDi DDIM	3000 epochs maximum; batch 256; jet attributes from test set	0.2800	0.0015	0.0010	0.590	0.0713	—
t	PC-JeDi EM	3000 epochs maximum; batch 256; jet attributes from test set	0.1500	0.0014	0.0012	0.590	0.0711	—
g	JetFUEL H-200K	Normalized tangent refs; evolving geometry	0.3745	0.0018	0.0013	0.565	0.0268	0 invalid
g	JetFUEL GQT-994K[g]‡	joint g/q/t; class-filtered FPND	0.1311	—	—	—	—	0.004% invalid
q	JetFUEL GQT-994K[q]‡	joint g/q/t; class-filtered FPND	0.2616	—	—	—	—	0.004% invalid
t	JetFUEL T-331K	H recipe; top specialist	2.8427	0.0058	0.0037	0.574	0.0566	0 invalid
t	JetFUEL GQT-994K[t]‡	joint g/q/t; class-filtered FPND	3.1530	—	—	—	—	0.004% invalid

‡ GQT-994K saves class-filtered FPND but joint-sample W1/Cov/MMD; those aggregate cells are intentionally blank. Its raw full-sample FPD is quarantined. PC-JeDi and MPGAN condition on jet attributes from the test set, while JetFUEL samples them in Stage 1.